RETURNS DISTRIBUTION

FINANCIAL ECONOMICS – SPRING 2026

PRE-CLASS WORK WITH CHATGPT

What is risk for an investor?

How do you model the distribution of returns?

What is the long-run evidence?

How do we predict future returns?

ARITHMETIC RETURNS

- We use the distribution of returns and not of asset prices

- The holding period return is the sum of the change in price between the initial and final moments of the period of holding the security plus any cash flow distributed as remuneration over the period. In the case of a stock (or index), the return is:

$$r_{t+1} = \frac{P_{t+1} - P_t + D_{t+1}}{P_t}$$

where P_t is the price at t, D_{t+1} the dividend. The return r_{t+1} is defined at t+1 to highlight that it will be known then. For a bond we consider the coupon:

$$r_{t+1} = \frac{P_{t+1} - P_t + C}{P_t}$$

LOG RETURNS

- +50%-50%=0 but invested wealth goes down 25%! To be able to sum return over time we need to use log returns, defined as $\log(\frac{P_{t+1}+FC_{t+1}}{P_t})$.

- If the price moves from 100 to 110 to 132, the log returns are $\ln(\frac{110}{100}) = 9.53\%$, $\ln(\frac{132}{110}) = 18.23\%$. The sum of the two log returns is $9.53\% + 18.23\% = 27.76\%$ that corresponds exactly to $\log(\frac{132}{100})$. The multi-period og returns is equal to the sum of the returns in each individual period.

- Differently from arithmetic returns, log returns do not aggregate across assets: the log return of a portfolio is not the weighted average of the log returns of the individual assets.

RETURNS AS RANDOM VARIABLES

- The future price and dividend are not known, but they can have various values

- In the event that we are able to determine the scenarios and the associated probabilities, we can describe the alternatives by means of a probability distribution

- The return is a random variable which can be describe ed through a probability distribution, in turn described by parameters such as expectation and standard deviation

THE NORMAL DISTRIBUTION

- Volatility is the standard deviation (or standard deviation) of the return distribution

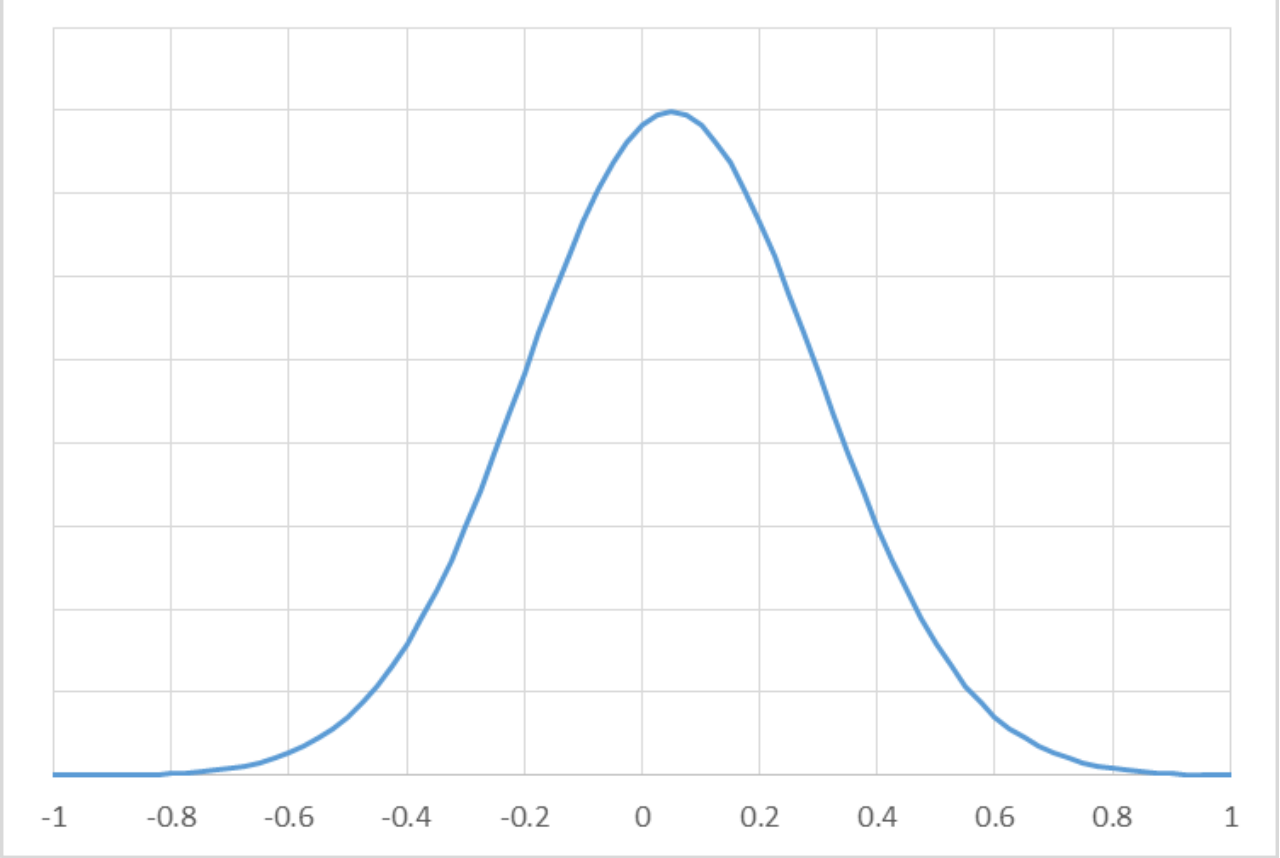
- The variance of the return is the square of the standard deviation

- In a "normal" distribution, the mean and standard deviation fully define the position of the probability distribution in the graph

The graph shows a normal distribution, with mean m=5% and volatility v=25%

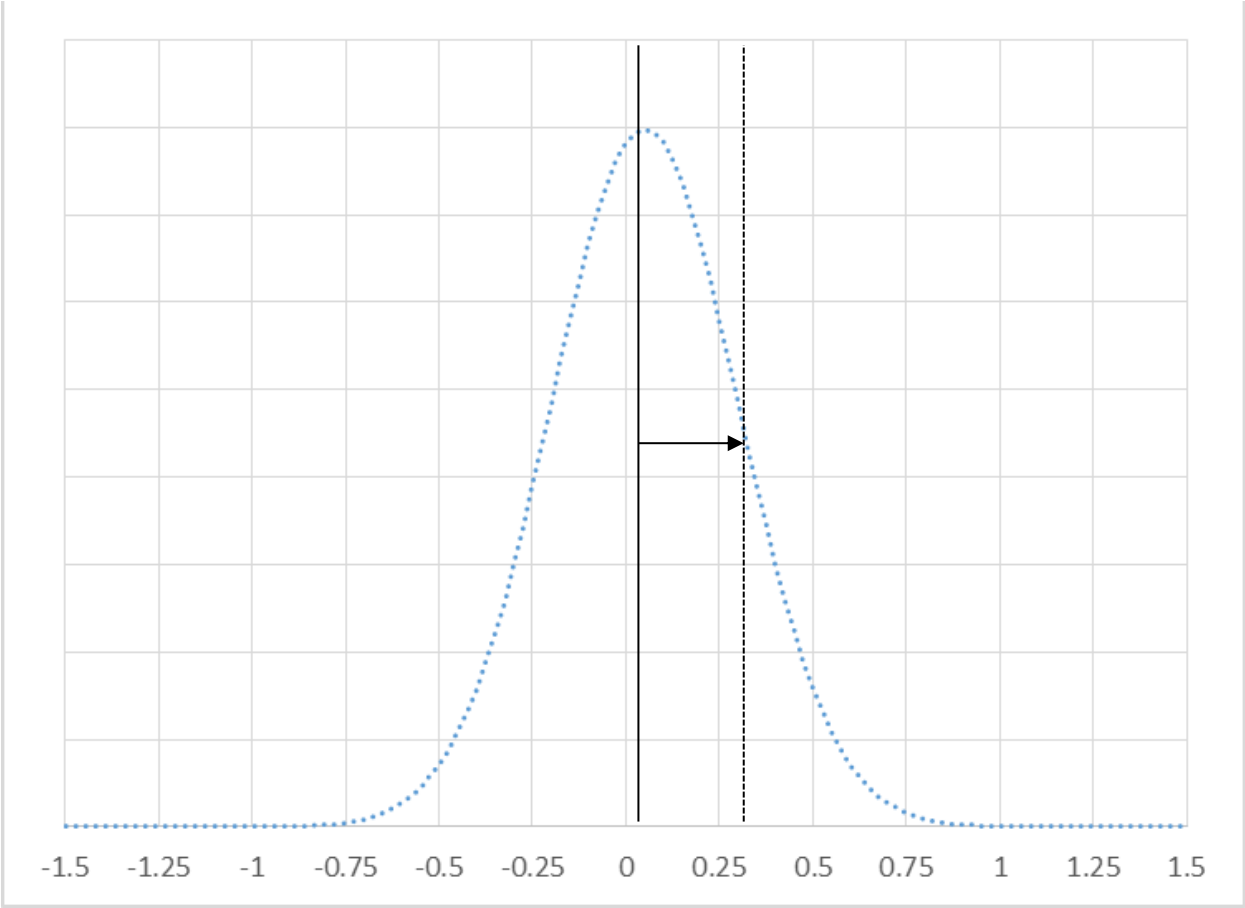
THE NORMAL DISTRIBUTION

The picture shows a normal distribution, with mean m=5% and volatility v=25%

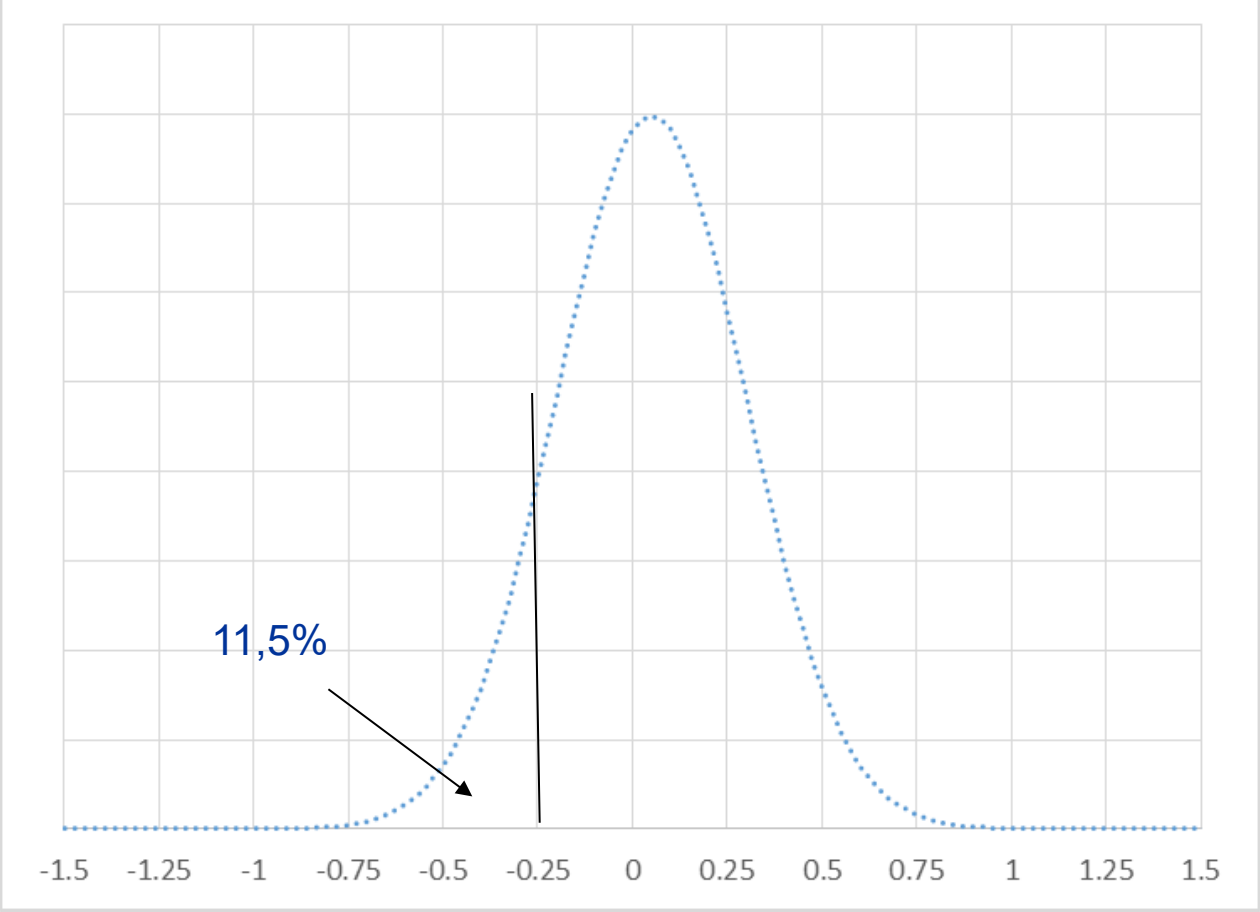


VOLATILITY

To draw the standard deviation consider the flexus



VOLATILITY AND RISK OF LOSS



QUESTION

- Compute the probability that return is less than -25% with mean 5% and volatility 25%

- Pr(R<-25%) = Pr((R−5%)/(25%) < (−25%−5%)/(25%)) = = Pr(Z<−1,2)

- From excel using =norm.dist(-1,2; 0; 1; 1) you get 11.5%. the same result is obtained by norm.dist(-25%; 5%; 25%; 1)

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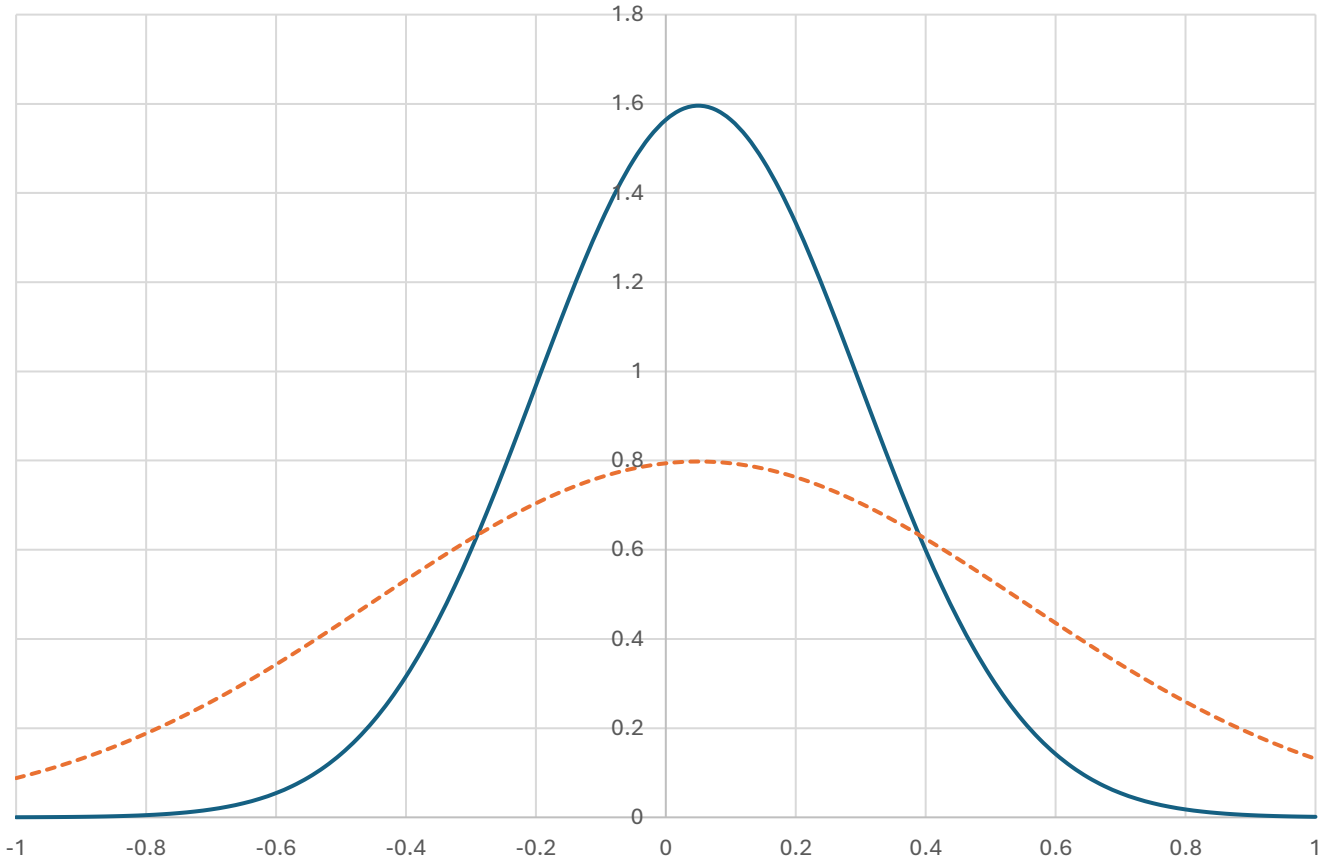
THE NORMAL DISTRIBUTION

Larger volatility describes more extreme events



THE NORMAL DISTRIBUTION

Relevant intervals

Interval: mean $\pm$	Probability that the return is included between:
$\pm\sigma$	68,26%
$\pm 2 \times \sigma$	95,44%
$\pm 3 \times \sigma$	99,74%

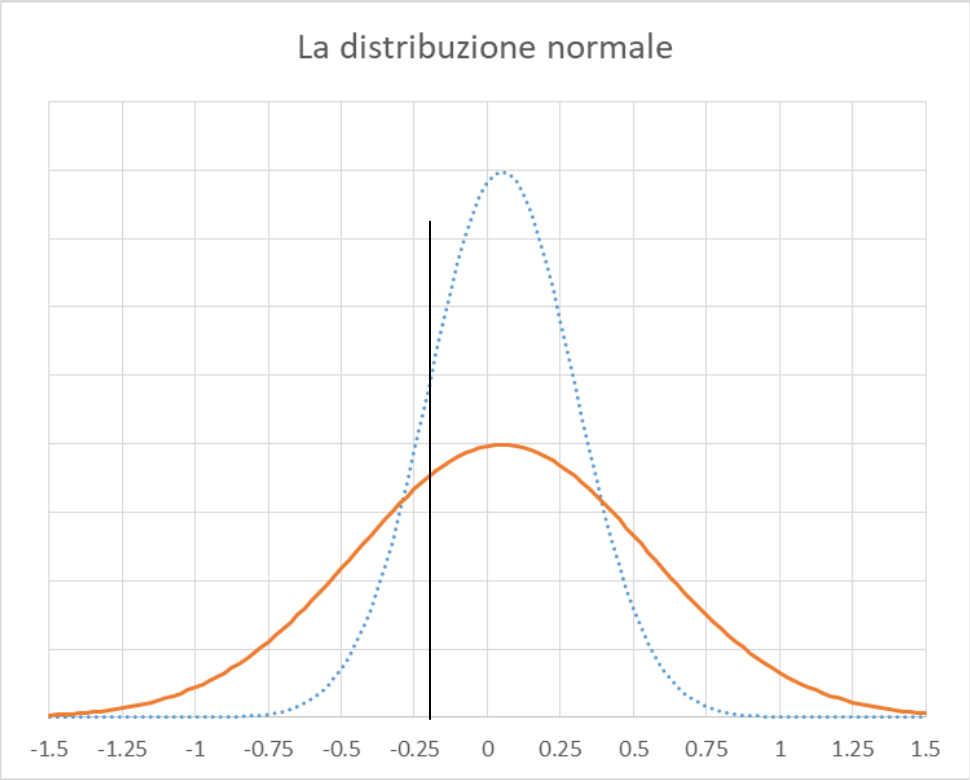
THE NORMAL DISTRIBUTION

- Two different normal distributions, both with mean m=5%, volatility is 25% and 50%

- Higher standard deviation increases the likelihood of much higher or lower than average returns

- When v=25%, the probability of having a return of less than -20% is 15.9%, while when v=50% the probability is 30.9%

- An increase in volatility with fixed mean increases the likelihood of losses so volatility can be interpreted as an indicator of loss risk

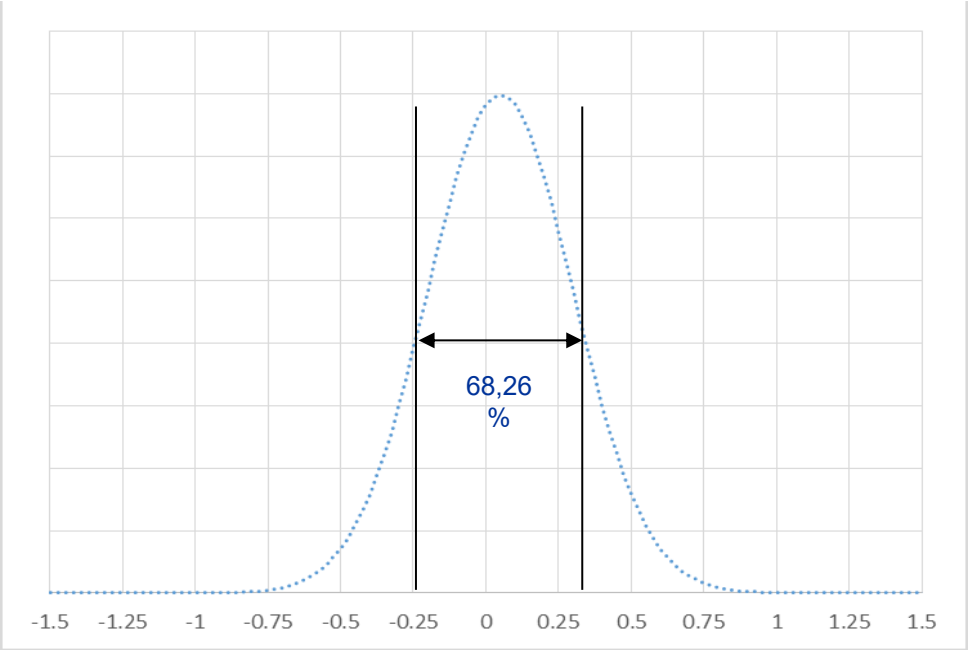


USEFUL INTERVALS

- With m=5% and v=25%, the probability that the return will be between average minus volatility and average plus volatility (“m−v; m+v”) is 68.26%

- The probability that the return is in the range “m−2×v; m+2×v” is 95.44%

- The probability that the return is in the range “m−3×v; m+3×v” is 99.74%



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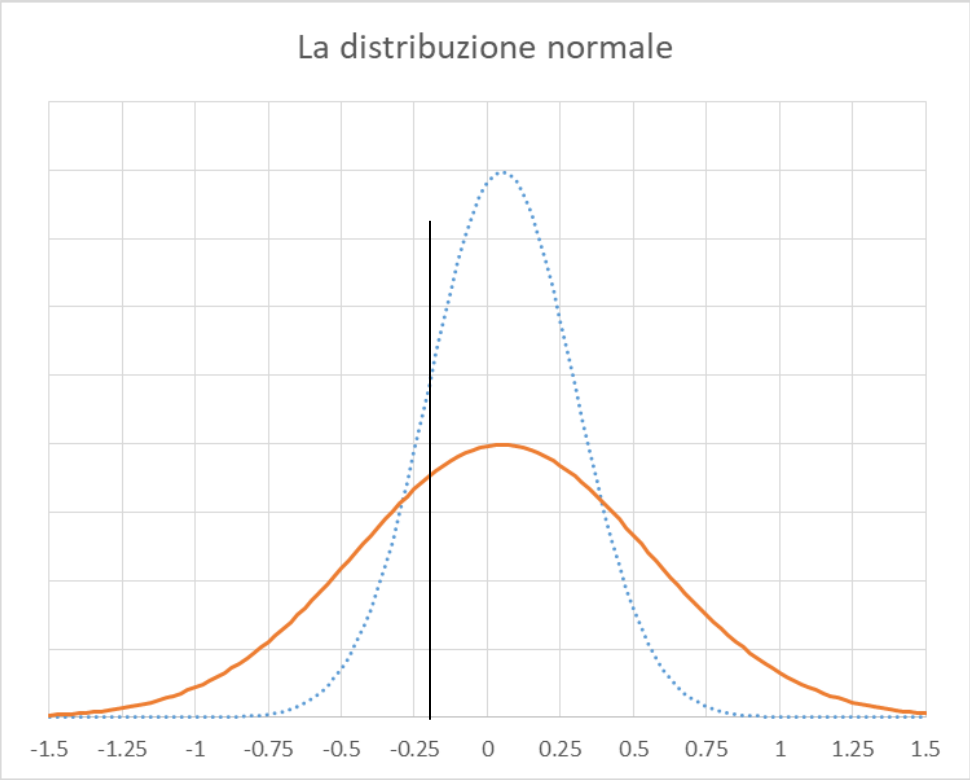


QUESTIONS

1: what is the probability that the return is between mean minus standard deviation and mean plus standard deviation when the mean=5% and volatility=20%?

2: what's the probability that the return is between +10% and -10% when the average is 5% and the volatility is 20%?

3: what's the probability that the return is between the average minus three times the standard deviation and the average plus three times the standard deviation when the mean=5% and volatility=20%?

ANSWERS

1: subtract from 1 twice the probability that the return is less than -15%, i.e. 1-2X norm.dist(-15%; 5%; 20%; 1) = 68.26%

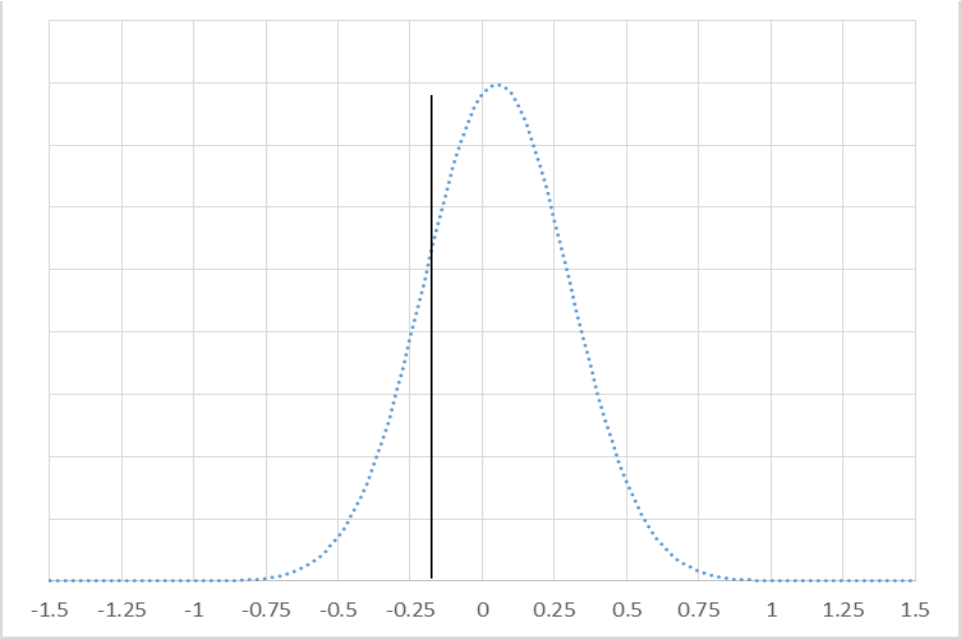
2: subtract from 1 twice the probability that the return is less than -10%, i.e. 1-2X norm.dist(-10%; 5%; 20%; 1) = 54.675%

3: subtract from 1 twice the probability that the return is less than -55%, i.e. 1-2X norm.dist(-55%; 5%; 20%; 1) = 99.73%.

FROM PROBABILITIES TO PERCENTILES

A risky asset has an expected return of 5% and volatility of 25%.

What is the minimum return that is achieved with 15% probability?

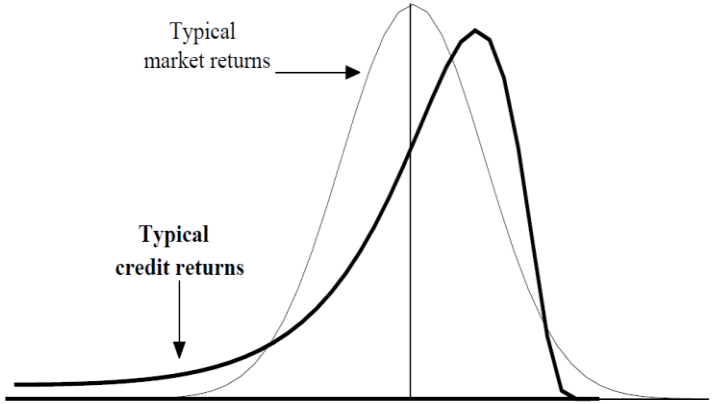


In statistical terms, we look for the value x such that Pr(r<x)=0.15

In excel NORM. INV(0.15; 0,05; 0.25)=-20.9%

NON-NORMALITY

For some asset classes, the normal does not work, for example to describe credit risk where the upside is limited by coupon payments while the downsi depends on the issuer's downgrades

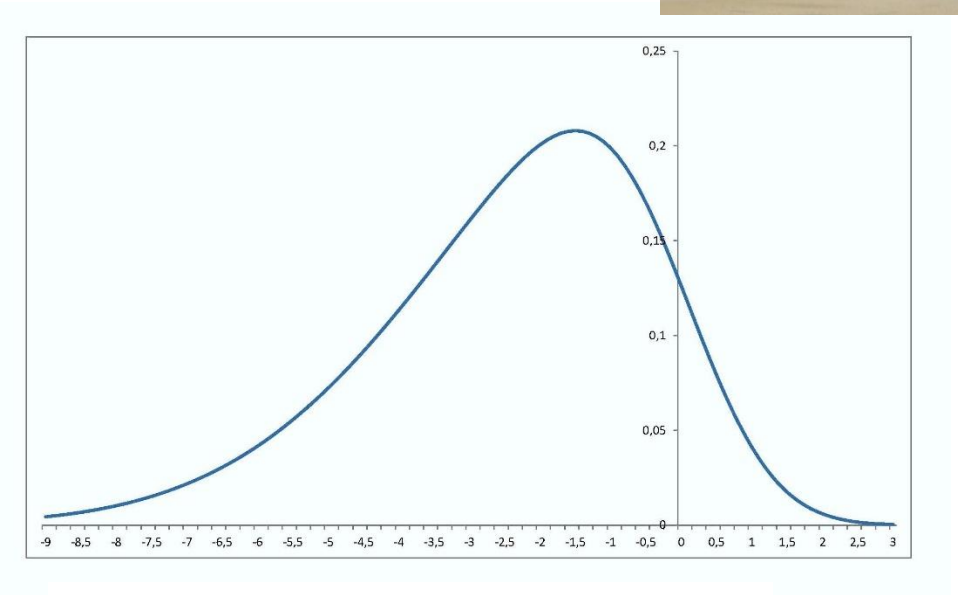


Looking at the long historical series of high-frequency stock returns, tails have a greater prominence than would be expected on the basis of the Gaussian, and other distributions, such as the t of student, with "thick tails" should be used

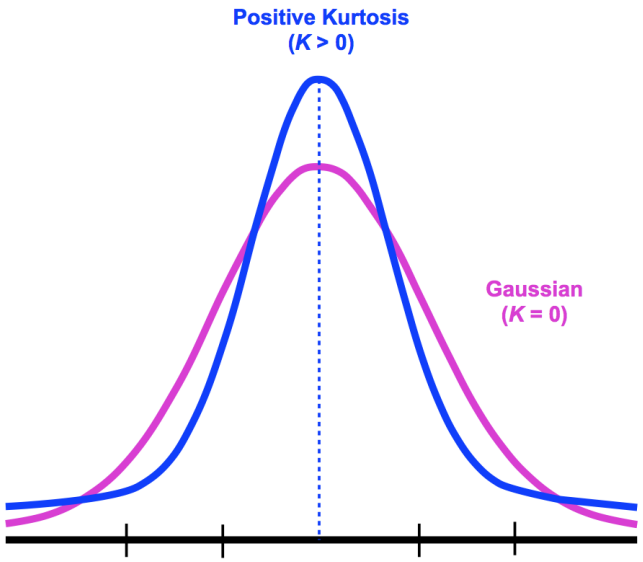
NON-NORMALITY

Skewness kurtosis

$$SK = \sum_{t=1}^{T} \frac{(r_t - \bar{r})^3}{T \times \hat{\sigma}^3}$$



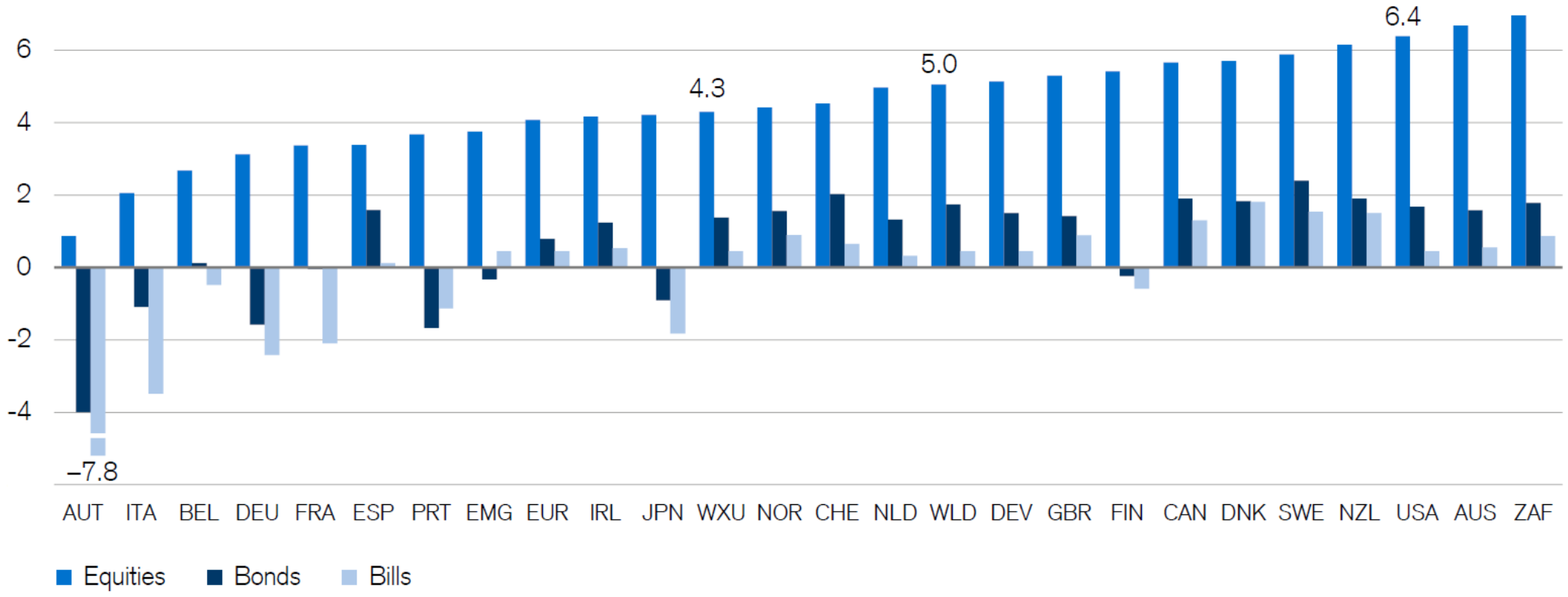
$$KU = \sum_{t=1}^{T} \frac{(r_t - \bar{r})^4}{T \times \hat{\sigma}^4}$$



Country/ Asset class	Geometric average (%)	Arithmetic average (%)	Standard deviation (%)	Minimum return (%)	Year of minimum return	Maximum return (%)	Year of maximum return
Equity							
Europe	4.1	5.9	19.7	-48.0	2008	75.2	1933
Japan	4.2	8.6	28.9	-85.5	1946	121.1	1952
United Kingdom	5.3	7.1	19.5	-56.6	1974	99.3	1975
United States	6.4	8.3	19.9	-38.6	1931	55.8	1933
World	5.0	6.5	17.4	-42.9	2008	67.6	1933

Country/ Asset class	Geometric average (%)	Arithmetic average (%)	Standard deviation (%)	Minimum return (%)	Year of minimum return	Maximum return (%)	Year of maximum return
Bond							
Europe	1.2	2.4	15.8	-52.6	1919	72.2	1933
Japan	-0.8	1.7	19.4	-77.5	1945	69.8	1954
United Kingdom	1.8	2.7	13.5	-29.9	1974	59.4	1921
United States	1.9	2.4	10.3	-18.1	1917	35.2	1982
World	1.9	2.5	11	-31.6	1919	46	1933

Figure 11: Real annualized returns (%) on equities versus bonds and bills internationally, 1900–2022



TEST FOR EQUALITY OF RETURNS

Assuming same volatility

$$\frac{(r_1 - r_{\mathrm{f}1}) - (r_2 - r_{\mathrm{f}2})}{\frac{\sigma}{\sqrt{T}}}$$

Notice: if the two average excess returns are 5% and 6% and volatility is 10% measured over 100 years then the test is equal to 1, not significant

Large volatility implies very long time series of data to assess statistical significance

EXPECTATIONS

- The expectation may be unconditional or conditional on the information available at the time. The unconditional expectation of return is that which is calculated by ignoring a specific set of information:

$$Er_{t+1} = \frac{EP_{t+1} - P_t + ED_{t+1}}{P_t}$$

- The conditional expectation is calculated using all the information available at a given time:

$$E_t r_{t+1} = \frac{E_t P_{t+1} - P_t + E_t D_{t+1}}{P_t}$$

EXPECTATIONS AND REGRESSIONS

- If we model data with a linear regression then we can use the estimated coefficients to compute the conditional expectation:

$$r_{t+1} = \alpha + \beta \times r_t + \varepsilon_{t+1}$$

$$E_t r_{t+1} = \alpha + \beta \times r_t$$

$$E r_{t+1} = \alpha / (1 - \beta)$$

R^2	0,0229
Adj. R^2	0,0183
Standard Error	0,0424
Numero oss.	215

	Coefficienti	*Standard Error*	*t Stat*	*P-value*
α	0,0039	0,0029	1,356	0,1765
β	0,1512	0,0676	2,235	0,0264

QUESTION

The intercept α=5% and β=0.1. In 2018, the market return was -60%.

1. What return forecast can be made for the long term?

2. And which one for the year 2019?

3. And for the year 2020?

4. How do you intuitively explain the difference between the forecast for 2019 and the forecast for the long term?

ANSWER

1) the long-term forecast (technically, the one obtained from the unconditional average, which ignores the latest information of 2018) is equal to the ratio between the intercept one minus the slope. We obtain 0.05 divided by 0.9, i.e. 5.5%.

2) The forecast for 2019 exploits the parameters estimated by the regression and the evidence for 2018 and is obtained by $0.05+0.1\times(-0.6)=-1\%$.

ANSWER

3) Forecast for 2020 is 0.05+0.1×(-0.01)=4.9%, very close to the long-term value.

$$E_t r_{t+1} = \alpha + \beta \times r_t$$
$$E_t r_{t+2} = \alpha + \beta \times E_t r_{t+1} = \alpha + \beta \times (\alpha + \beta \times r_t)$$

4) The long-term forecast is not influenced by the last return, while the one for 2019 must take into account the existence of a link between the returns of two consecutive years, measured by the correlation of the return of a certain year with the return of the following year. There is a weak autocorrelation in the data, and even with a strong shock of -50%, it can be expected that within a couple of years the return will return to its average values.

RETURNS, EXPECTATIONS, SHOCKS

The decomposition

$$r_{t+1} = E_t r_{t+1} + r_{t+1} - E_t r_{t+1} = r_{t+1} = E_t r_{t+1} + \varepsilon_{t+1}$$

shows that the total return can be interpreted as the sum of expectation and error

In 2008 markets dropped 30%, but is likely that at the beginning of the year investors expected a positive return, say 10%. If that was the case, the negative surprise was -40%